Implicit Differentiation

A Key Tool: The Chain Rule

$$\frac{d}{dx}\big(f(g(x))\big) = f'\big(g(x)\big)\cdot g'(x)$$

Implicit Differentiation

- **Consider the equation $xy + y = e^x$.** This defines y as a function of x. Find $y(x)$ and use it to find $\frac{dy}{dx}$.
  - **Solve explicitly for $y(x)$:** $y(x+1) = e^x \iff y = \frac{e^x}{x+1}$
  - **Differentiate via Quotient Rule:**

$$\frac{dy}{dx} = \frac{(x+1)\frac{d}{dx}[e^x] - e^x\frac{d}{dx}[x+1]}{(x+1)^2} = \frac{(x+1)e^x - e^x(1)}{(x+1)^2} = \frac{xe^x + e^x - e^x}{(x+1)^2} = \frac{xe^x}{(x+1)^2}$$

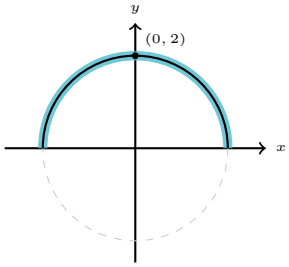
Now, consider the equation $x^2 + y^2 = 4$. What curve does this equation describe? Draw it!

- **Curve:** Circle of radius $r = 2$ centered at the origin $(0, 0)$.

Question: The equation $x^2 + y^2 = 4$ does not define y as a function of x. Why not?

- **Geometric Reason:** Doesn't pass the **Vertical Line Test (VLT)**. Vertical lines intersect the circle twice.
- **Algebraic Reason:** $y^2 = 4 - x^2 \implies y = \pm\sqrt{4 - x^2}$. Two outputs for many inputs (e.g., at $x = 1$, $y = \pm\sqrt{3}$).

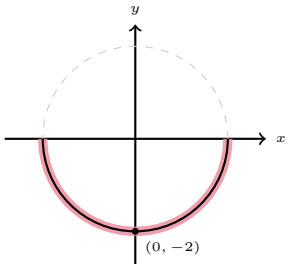
However, "near the point $(0, 2)$", y is a function of x. Highlight such a region on your graph above. What is this function? What is the derivative of this function at $x = 1$?



Top Hemisphere: $y = \sqrt{4 - x^2} = (4 - x^2)^{1/2}$
inner: $4 - x^2 \implies \text{inner}' = -2x$, outer: $u^{1/2} \implies \text{outer}' = \frac{1}{2}u^{-1/2}$

$$\frac{dy}{dx} = \frac{1}{2}(4 - x^2)^{-1/2}\cdot(-2x) = \frac{-x}{\sqrt{4 - x^2}} = -\frac{x}{y} \quad (\text{since } y = \sqrt{4 - x^2})$$

Do the same for the point $(0, -2)$ (and $x = -1$):



Bottom Hemisphere: $y = -\sqrt{4 - x^2} = -(4 - x^2)^{1/2}$

$$\frac{dy}{dx} = -\left(\frac{-x}{\sqrt{4 - x^2}}\right) = \frac{x}{\sqrt{4 - x^2}} = \frac{-x}{-\sqrt{4 - x^2}} = -\frac{x}{y} \quad (\text{since } y = -\sqrt{4 - x^2})$$

Notice that in both the above cases, across the entire circle, we can write:

$$\frac{dy}{dx} = -\frac{x}{y}$$

Implicit Differentiation

The Implicit Differentiation Technique

Even though $x^2+y^2=4$ does not define y as a function of x, it is still close to a function that the derivative makes sense, and the usual rules apply. Let's pretend y is a function of x, differentiate both sides, and solve for $\frac{dy}{dx}$:

- **Write $y=y(x)$:** $x^2+\big(y(x)\big)^2=4$
- **Differentiate both sides with respect to x:** $\frac{d}{dx}[x^2]+\frac{d}{dx}\left[\big(y(x)\big)^2\right]=\frac{d}{dx}[4]$
- **Chain Rule on the y-term:** $\frac{d}{dx}\left[\big(y(x)\big)^2\right]=2y(x)\cdot y'(x)$ (inner: $y(x)$, outer: $u^2 \implies 2u\cdot u'$)
- **Solve for $y'(x)$:**

$$2x+2y(x)\cdot y'(x)=0 \implies 2y(x)y'(x)=-2x \implies y'(x)=\frac{-2x}{2y(x)}=-\frac{x}{y}$$

Takeaway: Two lines of algebra, no square roots, and it works uniformly across the whole curve!

Geometric Significance & Why $\frac{dy}{dx}$ Depends on Both x and y

- Because the circle fails the vertical line test, a single x-value has multiple points on the curve:
  - At $(1,\sqrt{3})$ (top): $m=-\frac{1}{\sqrt{3}}$ (negative slope, tangent tilts downhill).
  - At $(1,-\sqrt{3})$ (bottom): $m=-\frac{1}{-\sqrt{3}}=\frac{1}{\sqrt{3}}$ (positive slope, tangent tilts uphill).
- **We also need the y-value** to determine which point on the curve we are evaluating!

Singularities: Vertical Tangent Lines

- **Are there any points where $\frac{dy}{dx}$ is not defined? What happens to the tangent line at these points?**
  - The derivative $\frac{dy}{dx}=-\frac{x}{y}$ is **undefined** when the denominator is zero: $y=0$.
  - Setting $y=0$ in $x^2+y^2=4 \implies x^2+0^2=4 \implies x=\pm 2$. Points: $(2,0)$ and $(-2,0)$.
  - **Geometric Meaning:** At $(2,0)$ and $(-2,0)$, the tangent lines are **vertical lines** ($x=2$ and $x=-2$).

Revisit: $xy+y=e^x$ via Implicit Differentiation

- **Question:** Let's revisit $xy+y=e^x$. What is $\frac{dy}{dx}$? Check that it's the same as what you got previously!
  - **Write:** $x\cdot y(x)+y(x)=e^x$.
  - **Differentiate via Product Rule on $x\cdot y(x)$:** $\frac{d}{dx}[xy(x)]=1\cdot y(x)+xy'(x)$.

$$(y+xy')+y'=e^x \implies xy'+y'=e^x-y \implies y'(x+1)=e^x-y \implies y'=\frac{e^x-y}{x+1}$$

  - **Check against explicit form:** Substitute $y=\frac{e^x}{x+1}$ into numerator:

$$y'=\frac{e^x-\frac{e^x}{x+1}}{x+1}=\frac{\frac{e^x(x+1)-e^x}{x+1}}{x+1}=\frac{xe^x+e^x-e^x}{(x+1)^2}=\frac{xe^x}{(x+1)^2} \quad \checkmark$$

Implicit Differentiation (Summary Statement)

What we carried out in these cases is a process of **implicit differentiation**:

1. We implicitly assume that y is a function of x ($y=y(x)$), making expressions like $y^2, y^3, e^y, \ldots$ compound functions: $y^2=[y(x)]^2, \quad e^y=e^{y(x)}, \ldots$
2. We then use the **Chain Rule** to differentiate them: $\frac{d}{dx}[y^n]=ny^{n-1}\cdot y', \quad \frac{d}{dx}[e^y]=e^y\cdot y'$.
3. We algebraically isolate all terms with y' and solve for $\frac{dy}{dx}$.